End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Course Name: B.Tech CSE/ECE/IT/MAE/DMAM, Semester: II
(May, 2024) (2022 onwards)

Subject Code: BAS 104	**Subject: APPLIED PHYSICS II**
Time :3 Hours	**Maximum Marks :60**

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(5*4 =20)**	
	(a) Show that, in free space EM waves are transverse in nature.		5
	(b) Calculate the Fermi energy of electrons at $0K$ for a metal with electron density of $10^{28}m^{-3}$.		3
	(c) A rod placed in a frame of reference is moving with a velocity $0.6c$ in a direction parallel to its length. Calculate the percentage of contraction in its length.		5
	(d) What are the key characteristics that define a good sensor? Discuss the importance of each characteristic in ensuring the sensor's functionality and reliability in various applications.		5
	UNIT-I		
Q2	(a) Starting with Maxwell's Equations, derive the electromagnetic wave propagation equation for a conducting medium. Taking plane monochromatic EM wave as its solution, explain the characteristics of its propagation vector and phase profiles of the resultant electric and magnetic fields. (b) Discuss the role of the Poynting vector in conservation of energy in electromagnetic systems.	**(5,5)**	9
Q3	(a) Using Maxwell's equations, obtain the electromagnetic wave equations in an isotropic dielectric medium and show that the speed of wave is less than its speed in vacuum. (b) Find the expression for the magnetic field that induces the given electric field: $\vec{E} = k[yz\hat{\imath} + 3z\hat{\jmath} + 4y\hat{k}]\cos(\omega t)$	**(5,5)**	
	UNIT-II		
Q4	(a) Derive the expression for Fermi Energy level for p-type semiconductor. Draw the resultant energy level diagram. (b) Find out reciprocal lattice vectors for a space lattice defined by the following primitive translation vectors: $\vec{a} = 5\hat{\imath} + 5\hat{\jmath} - 5\hat{k}$ $\vec{b} = -5\hat{\imath} + 5\hat{\jmath} + 5\hat{k}$ $\vec{c} = 5\hat{\imath} - 5\hat{\jmath} + 5\hat{k}$ where $\hat{\imath}$, $\hat{\jmath}$ and $\hat{k}$ are the unit vectors along x, y and z axes.	**(5,5)**	
Q5	(a) Discuss the Kronig-Penney model for the motion of electrons in periodic potential. Show from $E-k$ diagram that the materials can be classified into conductors, insulators and semiconductors. (c) A semiconductor contains acceptors and the hole concentration is given equal to $10^{22}m^{-3}$. Assume that the density of states in the valence band $N_v = 10^{25}m^{-3}$. How far above the valence band in eV does the Fermi Energy Level lie at $300K$?	**(5,5)**	10

	(Take Boltzmann's constant, $K = 8.617 \times 10^{-5} eV\ K^{-1}$)		
	UNIT-III		
Q6	(a) Starting from Lorentz transformation equations for space and time coordinates, derive equations for relativistic addition of velocities. (b) A proton from outer space is moving towards earth with velocity $0.99c$ as measured in earth's frame. A spaceship, travelling parallel to the proton, measures proton's velocity to be $0.97c$. Find the approximate velocity of the spaceship in the earth's frame of reference.	(5,5)	10
Q7	(a) Derive the Lorentz transformations for the coordinates of space and time between two inertial frames, one of which (S' Frame) is moving at a constant velocity v relative to the other (S Frame)? Compare the Lorentz transformations with the Galilean transformations of classical mechanics, and explain how these transformations alter our traditional concepts of time and space? (b) Two events E_1 and E_2 take place in an inertial frame S with respective space, time coordinates (in S.I. units): $E_1(t_1 = 0, x_1 = y_1 = z_1 = 0)$ and $E_2(t_2 = 0, x_2 = 10^8, y_2 = 0, z_2 = 0)$. Another inertial frame S' is moving with respect to S with a velocity $\vec{v} = 0.8c\,\hat{\imath}$. Find the time difference $(t_2' - t_1')$ as observed in S'.	(7,3)	
	UNIT-IV		
Q8	(a) A beam of X-rays is incident on a sodium chloride crystal with lattice spacing 0.282nm. The first order Bragg reflection is observed at a glancing angle of $8°35'$. What is the wavelength of X-rays? At what angle, would the third order Bragg's reflection occur? (b) Explain the production and detection of X-rays. Give three application of X-rays.	(5,5)	
Q9	(a) Describe the construction and working principle of a Light Dependent Resistor (LDR). Discuss a potential application of LDRs in electronic circuits and the impact of light intensity on this application. (b) Differentiate between the following: (i) Passive and Active sensors (ii)Sensitivity and Accuracy	(7,3)	10